



Effects of Workload on Radiation Vulnerability of RISC-V System-on-Chip

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COTS: Commercial-Off-The-Shelf

CS: Cross Section

DDR: Double Data Rate

ECC: Error Correcting Code

MM: Matrix Multiplication

RAM: Random Access Memory

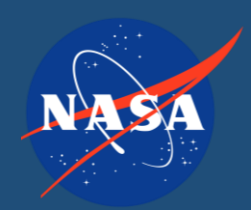
SEDED: Single Error Correction, Double Error Detection

SEU: Single-Event Upset

SEFI: Single-Event Functional Interrupt

SoC: System-on-Chip

TLYF: Test Like You Fly



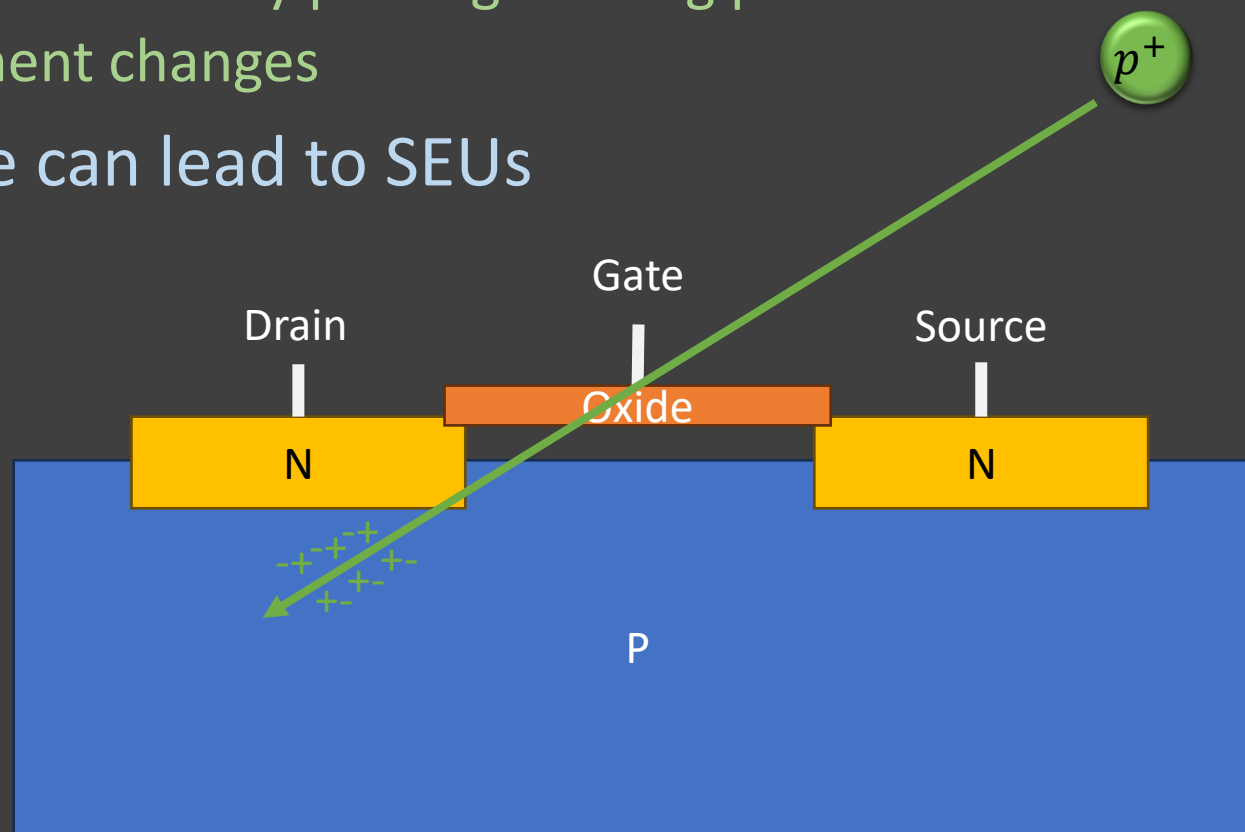
Acknowledgements

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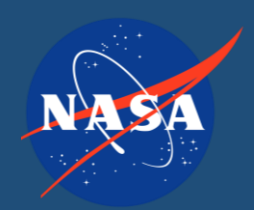


Processor Single-Event Effects

- Ionizing Radiation causes errors in hardware which propagate to software
 - Creation of electron-hole pairs generated by passing ionizing particles
 - Transient and potentially permanent changes
- Faults in the registers and cache can lead to SEUs
 - Silent data errors
 - Segmentation Faults
 - **Kernel panic**



SEU: Single-Event Upset

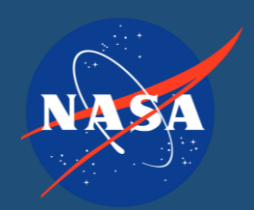


Problem Statement

- SoCs are complex
 - Hard to test
 - Hard to isolate complications and subsystems
- Current Fix: “Test as you Fly” or “Test Like You Fly” (TLYF)
 - Test the **exact** configuration and software as is going to be on mission
 - *Any* change, even compilation/optimization flags, could invalidate tests
- Need to find and standardize experimental controls
 - Control and isolate experimental variables to observe effects
 - Find correlations between isolated subsystems and reliability



SoCs: Systems-on-Chip



Linux Memory Spaces

- Linux system memory in two key parts
 - User space
 - Kernel space
- Hardware drivers and OS code exist in kernel space
 - Necessary to bridge the gap between hardware and OS
 - Huge part of the kernel are drivers
- Driver crash or failure can lead to kernel panic
 - Hangs entire system
 - **⇒ Drivers leading cause of significant kernel failures**

	Address	Data
KERNEL SPACE	0000 0000	YYYY YYYY
	0000 0001	YYYY YYYY
	...	...
	XXXX XXXX	YYYY YYYY
USER SPACE	XXXX XXX(X+1)	YYYY YYYY
	...	...
	...	...
	...	...
	...	...
	...	...
	FFFF FFFF	YYYY YYYY

*simplified for illustration

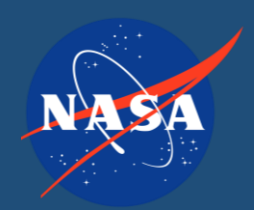


Previous Tests

- Proton irradiation of TUL PYNQ-Z2
 - ARM Cortex-A9 processor
 - **Parity ECC in cache**
- Found discrepancies between SEFI frequencies under different workloads
 - Idling crashed more often than mathematical workload
 - Unclear if statistical phenomenon
- As always, want more data



ECC: Error-correcting code
SEFI: Single-event functional interrupt



Linux Setup

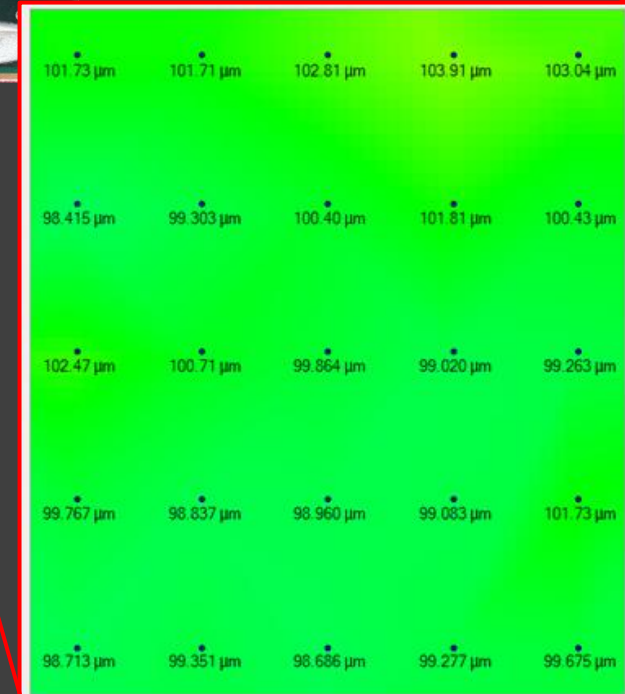
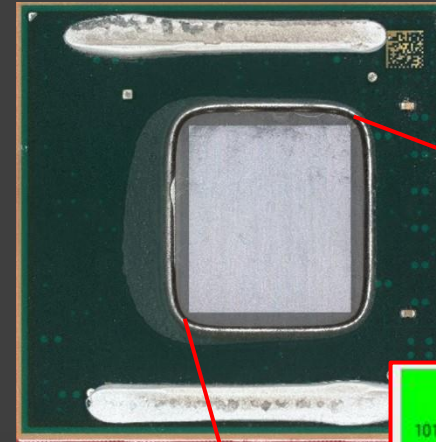
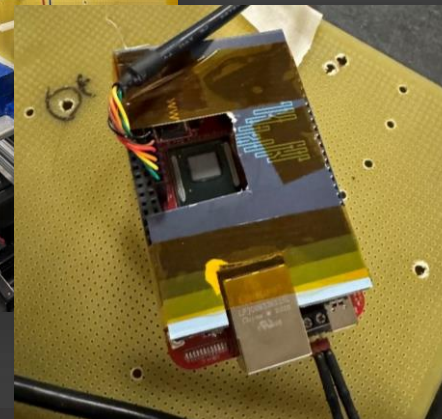
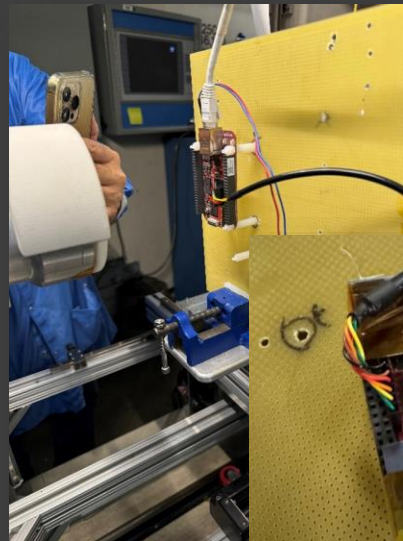
- BeagleBoard Beagle-V Fire
 - 4x RISC-V RV64GC cores
 - **Both L1 and L2 caches have SECEDED**
- Generate Buildroot Linux images for **2 configurations**
 - Ensure minimalist Linux image with as many parts known as possible
 - IDLE configuration and Matrix Multiplication (MM) configuration
 - First stage bootloader was PolarFire SoC Hart Software Services
- Initramfs
 - Load entire image in RAM from eMMC
 - Refresh each reboot to fresh image.
 - Operations and driver setup run with initscripts
- Focus on crashes and kernel panics



SoC: System-on-Chip
eMMC: embedded MultiMediaCard
SECEDED: Single-Error Correction,
Double-Error Detection

Test Setup

- System needs to be thinned
 - Thinned to $100\text{ }\mu\text{m}$ by Sage Analytical
- Shield made from Cardboard to protect flash memory from radiation
 - Focusing on only the processing cores
- Lawrence Berkeley National Labs Heavy Ion Facility
 - $20\frac{\text{MeV}}{\text{amu}}$ beam tune
 - Mounted $\sim 1.75\text{ cm}$ from port





Software Configurations

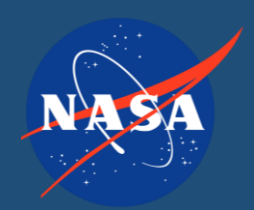
- IDLE
 - System set to idle using “sleep” bash command
 - Act as a mode with little memory transfers to and from cache
 - Only critical kernel/driver operations active
- MM
 - 500 × 500 naïve matrix multiplication implemented in C
 - Act as a higher workload with many data transfers needed for calculation

$$\begin{bmatrix} A \end{bmatrix} * \begin{bmatrix} B \end{bmatrix} = \begin{bmatrix} C \end{bmatrix}$$

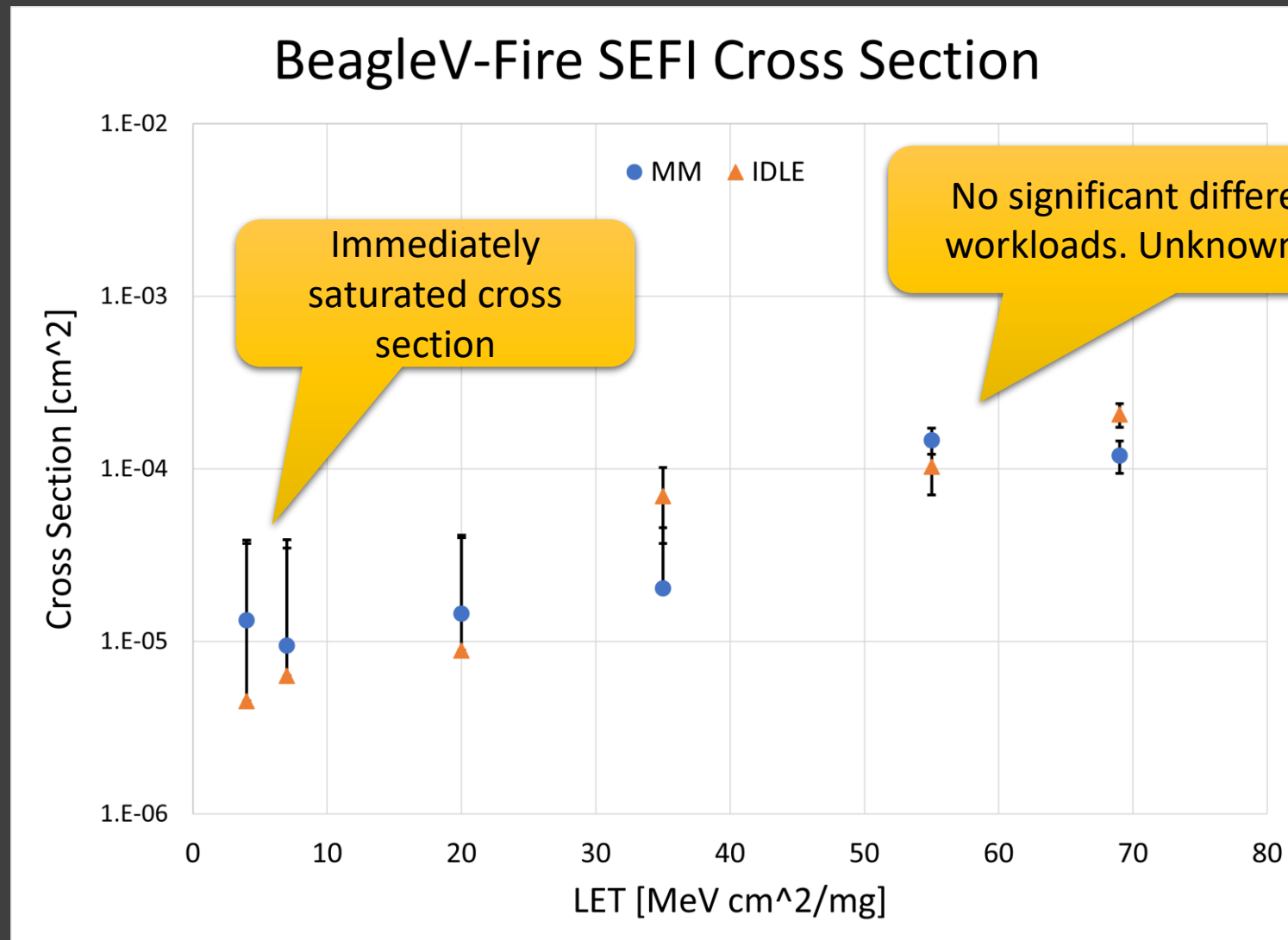


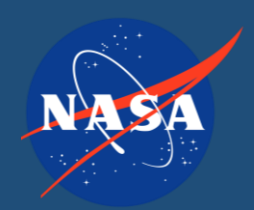
Error Signatures

- Four errors classified and reported by SECDED implementation:
- DataError
 - A correctable error in the data, such as only a single bit flip
- DataFail
 - An uncorrectable error in the data, such a multi-bit upset
- DirError
 - A correctable error in the cache coherence directory used for cache synchronization between processor cores
- DirFail
 - An uncorrectable error in the cache coherence directory

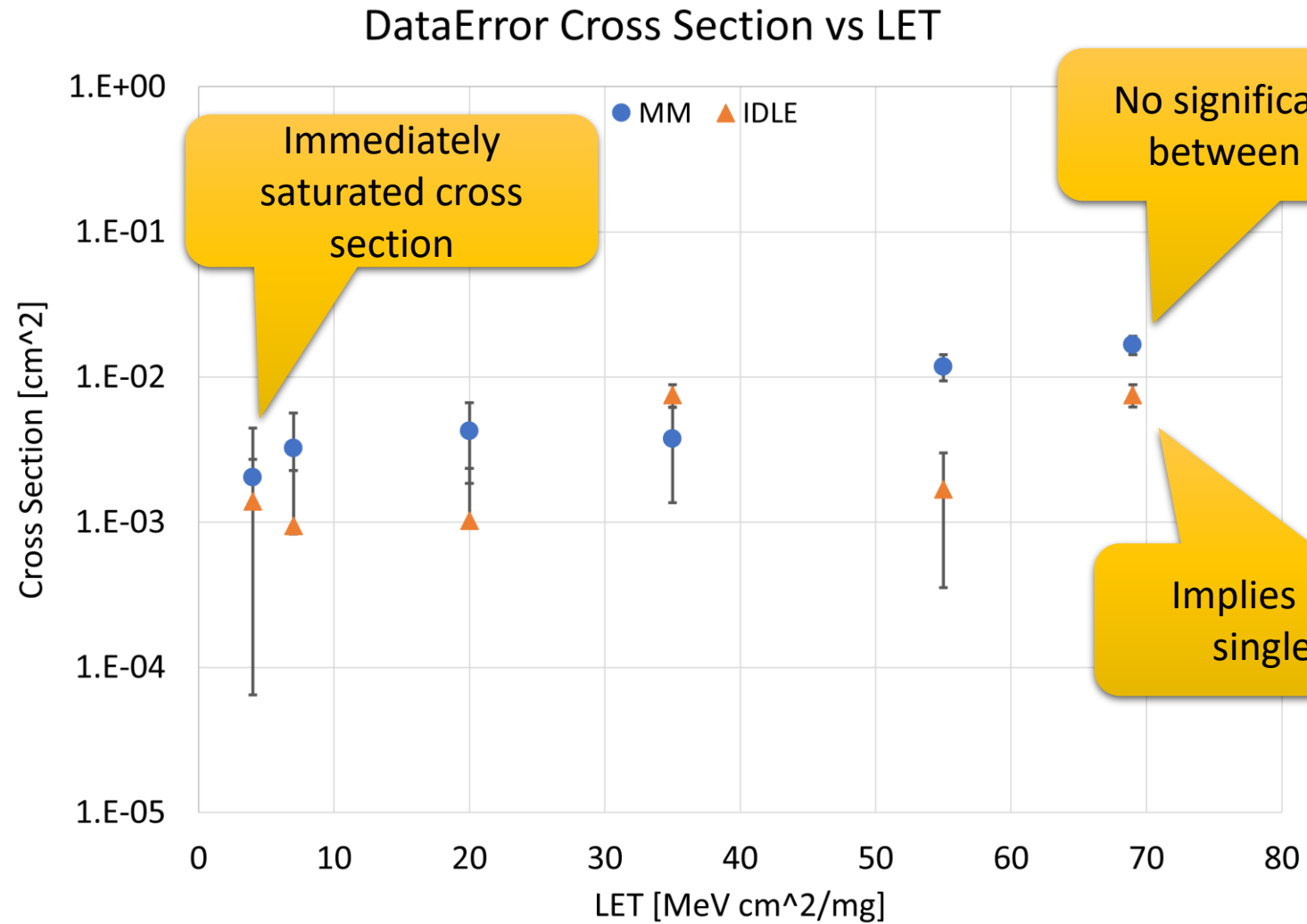


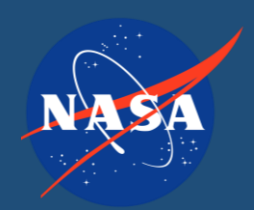
SEFI Cross Section



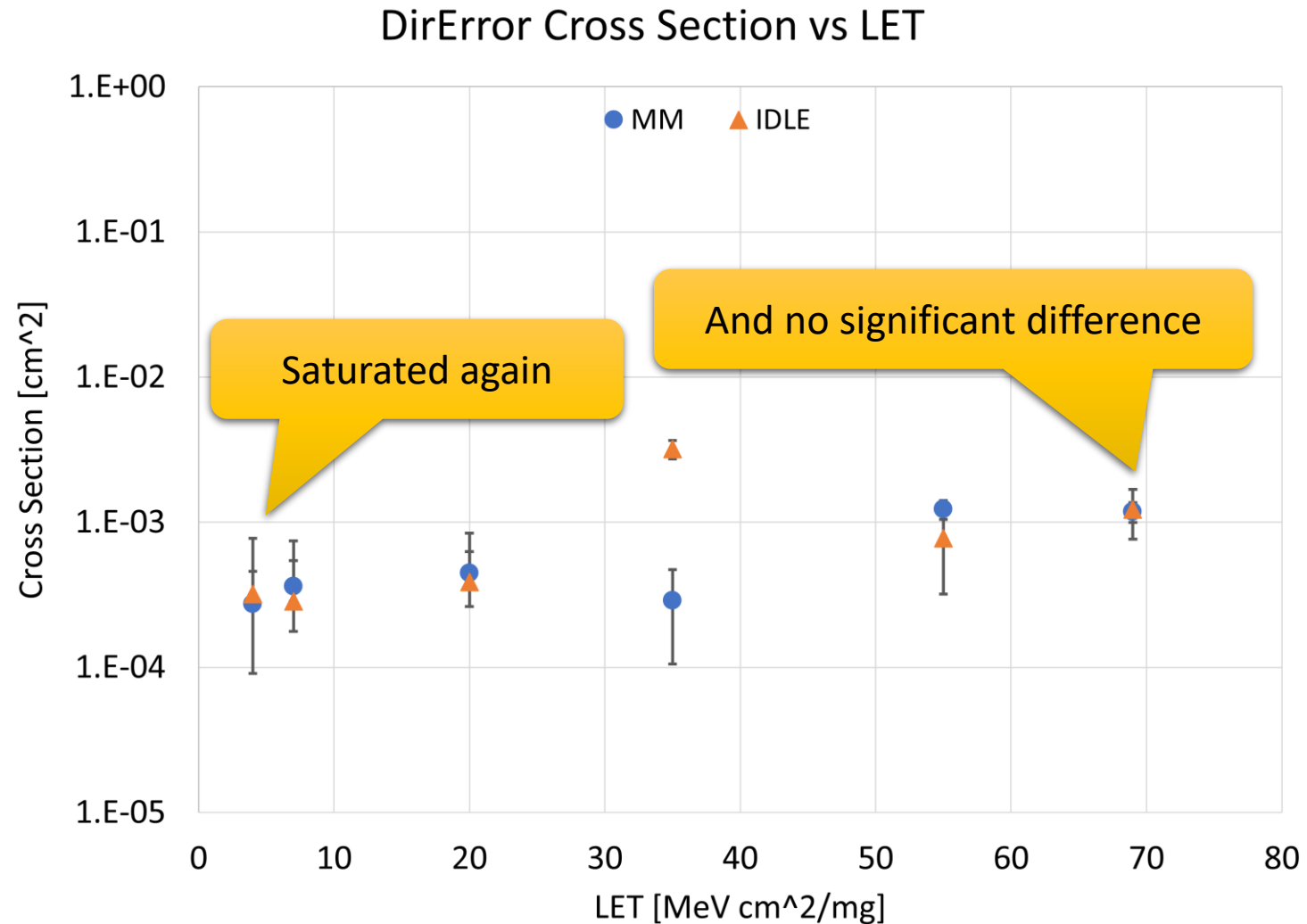


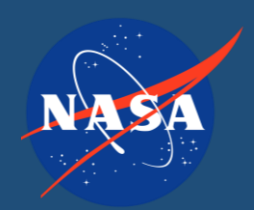
DataError





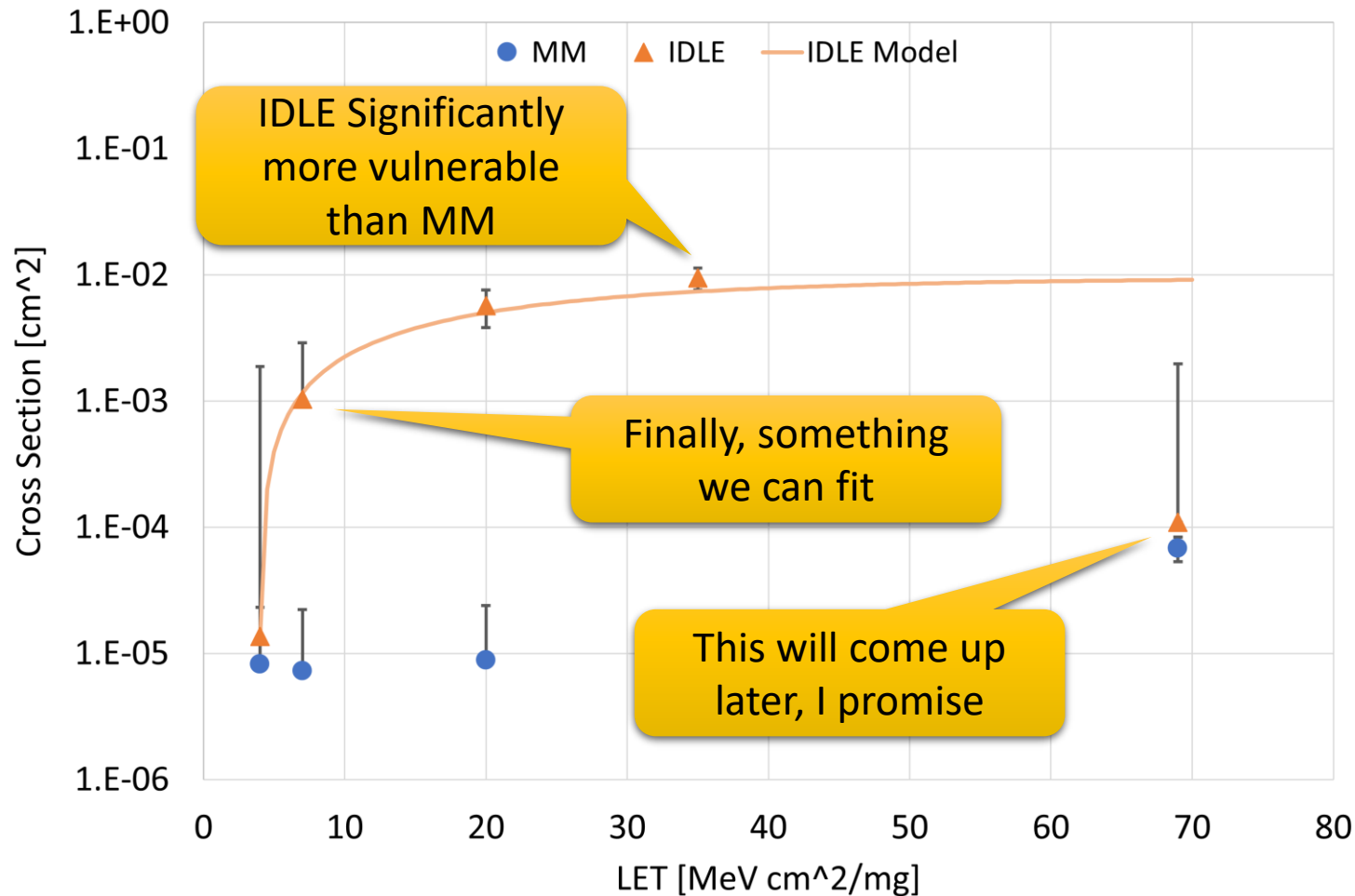
DirError





DataFail

DataFail Cross Section vs LET



Model Parameter	Fitted Value
Onset LET	$3.96 \text{ MeV} \frac{\text{cm}^2}{\text{mg}}$
Saturated CS	$9.471\text{E-}3 \text{ cm}^2$
Shape	1.052
Width	$20.98 \text{ MeV} \frac{\text{cm}^2}{\text{mg}}$

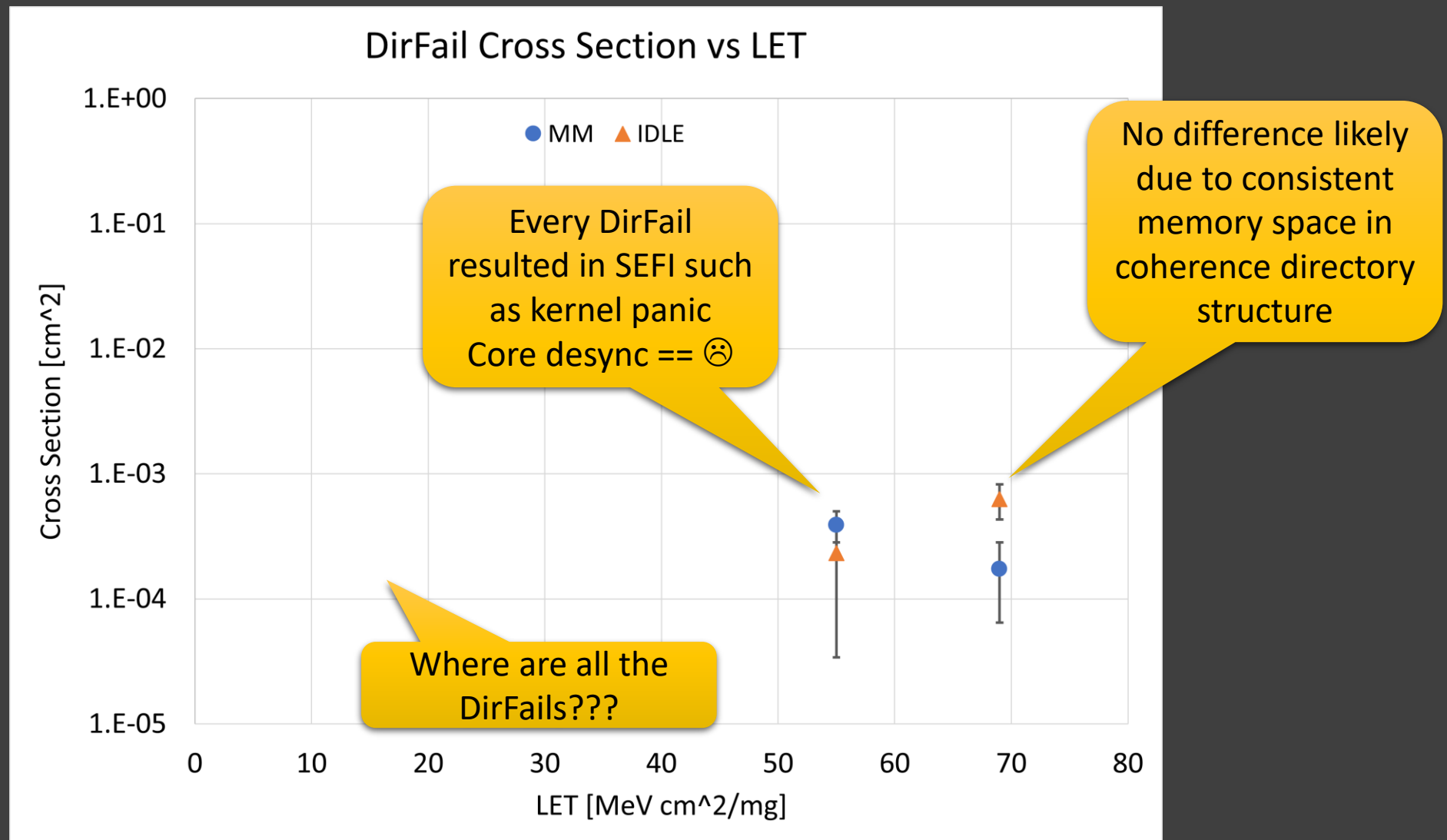
IDLE sees more multi-bit upsets in cache than matrix-multiplication mode

Likely due to memory movement

More data being pulled/refreshed from non-radiated DDR for MM as calculation happens.

Meanwhile, memory in IDLE sits in cache and waits for errors to build up

CS: Cross Section





Conclusions

- No differences in SEFI results between both workloads
 - Differs from PYNQ proton test results
 - Could be due to statistical insignificance in proton test
 - Could also be due to differences in ECC (Parity in PYNQ vs SECDED in BeagleV)
 - Implies static memory space where error $\Rightarrow$ SEFI
- No differences in workload in correctable errors, both data and coherence
 - i.e. the SECDED ECC agnostic about the workload
- Cache Coherence failures consistent between workloads
 - Likely due to core synchronization memory space static regardless of workload
- Data Failures significantly worse in IDLE case than MM
 - Data sitting around in cache for longer $\Rightarrow$ higher likelihood of multiple error accumulation
 - Multiple error accumulation $\Rightarrow$ SECDED ECC fails
- Therefore, workload **does** have some affect on reliability, depending on ECC



Future Work

More data, please!!

More architectures

More ECC implementations

More board designs

Replication experiments

Want to design better experimental controls and optimize
Test Like You Fly workloads

Thank you for your time! Questions? Comments? Philosophies?